Python Data Analyst Specialist

(Founder’s Mentoring Program)

Executive Retail Intelligence Hub Dashboard: Project Report

November 7, 2026

Table of Contents

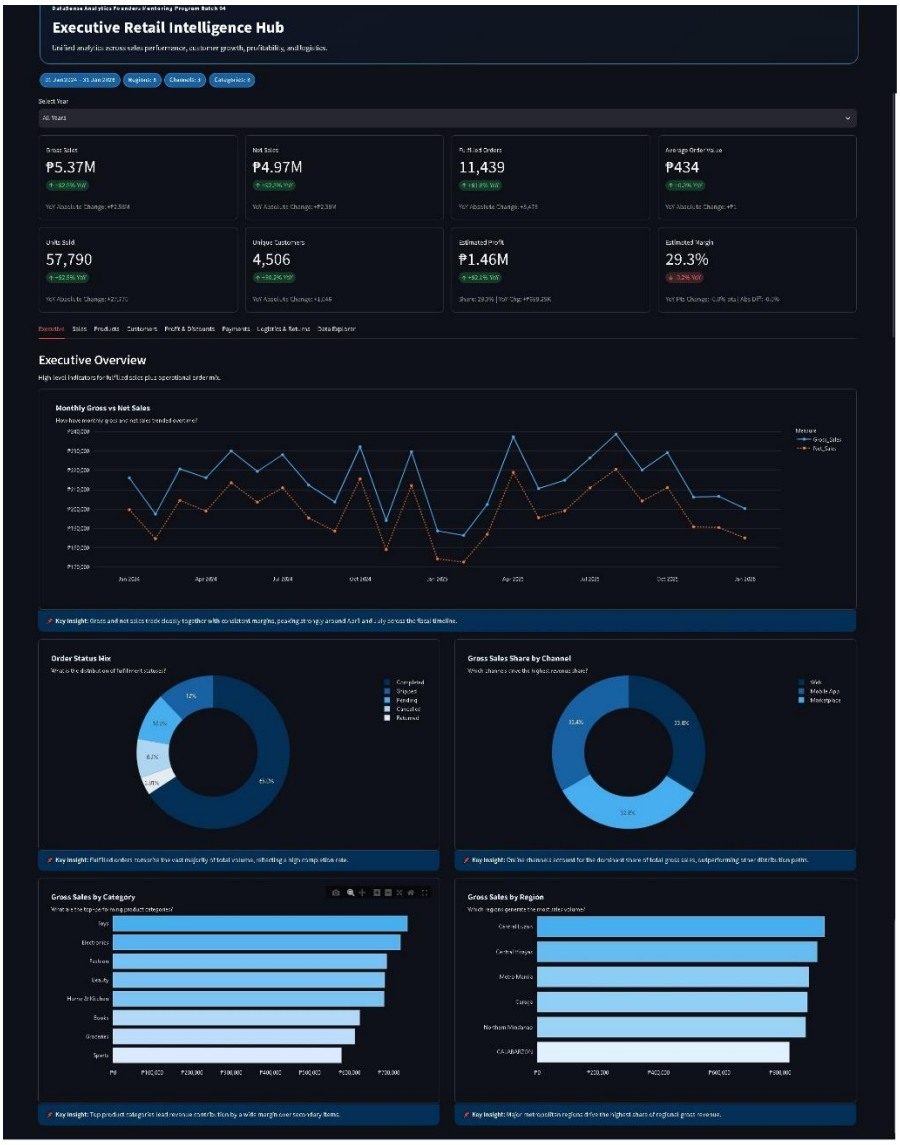
Section 1: Introduction 3
Background 3
Section 2: Dashboard....3
Section 3: Project Overview 4
Objectives 4
Scope 4
Purpose 4
Section 4: Data Analysis & Methodology 5
4.1 Data preparation (load_data()) 5
4.2 Key metrics and calculations 5
4.3 Filters 6
4.4 Analytical techniques 6
4.5 Executive snapshot (All Years) 6
Section 5: Key Takeaways 6
Section 6: Insights 7
Section 7: Summary 8
Section 8: Conclusions 8
Section 9: Recommendations 8
Now 8
Next 8
Later....9

Section 1: Introduction

Background

Retail businesses generate data in many separate places: orders, products, customers, payments, shipments and returns. When these sit in separate files, it is hard to answer simple executive questions such as "Are sales growing?", "Which products and regions perform best?", "Are we making money after discounts?" and "How many orders actually reach the customer?"

Section 2: Dashboard



Section 3: Project Overview

Item	Details
Project name	Executive Retail Intelligence Hub
Program	DataSense Analytics Founders Mentoring Program, Batch 04
Author	Ranie Roldan Cruz
Type	Interactive business intelligence dashboard with written analysis
Business domain	Retail / e-commerce (Philippines, amounts in ₱)
Data period	01 Jan 2024 – 31 Jan 2026
Tech stack	Python, pandas (data cleaning and metrics), Plotly Express (charts), Streamlit (web app)

Objectives

- Combine seven separate retail tables into one clean, trusted data model.
- Track the core business KPIs (sales, orders, customers, profit and margin) with year-over-year comparison.
- Break performance down by time, product, customer, region, channel, discount, payment and logistics.
- Give executives a quick read on where the business is strong, where it leaks revenue, and what to do next.

Scope

- **Data:** seven related tables: customers, products, orders, order_items, payments, shipments, returns
- **Period:** 01 Jan 2024 – 31 Jan 2026
- **Coverage:** 6 regions, 3 sales channels, 8 product categories (amounts in Philippine pesos, ₱)
- **Dashboard:** 8 tabs (Executive, Sales, Products, Customers, Profit & Discounts, Payments, Logistics & Returns, Data Explorer) with year and sidebar filters

Purpose

The **Executive Retail Intelligence Hub** was built as the Batch 04 project of the DataSense Analytics Founders Mentoring Program. It turns seven raw retail CSV files into a single, cleaned and interactive Streamlit dashboard that brings together sales performance, customer growth, profitability, payments, and logistics in one place.

Section 4: Data Analysis & Methodology

The analysis follows five steps: clean → join → calculate → filter → visualise and compare.

4.1 Data preparation (load_data())

- De-duplication: exact duplicate rows are dropped in every table.
- Standardisation: gender and product category labels are mapped to one spelling (e.g. Fashon → Fashion, Elec → Electronics); money fields have ₱ and commas stripped; mixed-format dates are parsed.
- Validation: ages outside 18–100 become blank; discounts outside 0–1 become 0; order items with non-positive quantity, missing price, or unknown product/order are removed; orders with unknown customers are removed (referential integrity).
- Joins: order items are merged with orders, products and customers (many_to_one validated) into an item-level sales table (sales_all), then rolled up to an order-level table (order_summary).
  Data-quality check: the Data Explorer tab compares raw vs clean row counts per table.

4.2 Key metrics and calculations

Metric	Calculation
Gross Sales	quantity × unit_price_at_purchase
Discount Amount	Gross Sales − Net Sales
Net Sales	Gross Sales × (1 − discount_pct)
Estimated Cost	quantity × unit_cost
Estimated Profit	Net Sales − Estimated Cost
Estimated Margin	Estimated Profit ÷ Net Sales
Average Order Value (AOV)	Mean net value per order
Units Sold / Fulfilled Orders / Unique Customers	Sum of quantity / distinct order_id / distinct customer_id
Delivery Days	delivered_date − ship_date (negative values blanked)
Return Rate	Orders with a return ÷ orders in scope
Fulfilled	Orders with status **Completed** or **Shipped** (used for sales KPIs)

4.3 Filters

- Year selector (All Years default) controls the KPI strip and the Executive charts.
- Sidebar filters: order date range, region, sales channel, product category, shipping method, with a Reset Filters button.
- Scope rules: sales and profit use fulfilled orders only; order status mix, payments, shipments and returns use all orders in scope.

4.4 Analytical techniques

- Aggregation and grouping by month, weekday, hour, category, brand, region, city, channel, loyalty tier and courier
- Time-series trend analysis (monthly gross vs net, revenue/profit/discount)
- Period-over-period comparison (YoY % change, absolute change, margin point change) Share-of-total and ranking (donut charts, ranked bars)
- Distribution analysis (order size, customer spend brackets, discount rates, delivery days) Segmentation (loyalty tier, region, category, sales channel)
- Operational and quality analysis (payment failure, shipment status, returns and refunds, raw vs clean rows)

4.5 Executive snapshot (All Years)

KPI	Value	Derived
Gross Sales	₱5.37M	
Net Sales	₱4.97M	≈ ₱0.40M discounts (≈ 7.4% of gross)
Fulfilled Orders	11,439	
Average Order Value	₱434	
Units Sold	57,790	≈ 5 units per order, ≈ ₱86 net per unit
Unique Customers	4,506	≈ 2.5 orders per customer
Estimated Profit	₱1.46M	
Estimated Margin	29.3%	

Section 5: Key Takeaways

- The headline growth is misleading. The "+92% YoY" shown on Gross Sales, Net Sales, Orders and Units is caused by comparing 25 months against a prior window containing only 13 months of data (25 ÷ 13 ≈ 1.92).
- Sales are flat to slightly declining. Monthly gross sales stay between ≈ ₱186K and ≈ ₱239K (≈ ₱215K average), Jan 2026 (≈ ₱200K) is below Jan 2024 (≈ ₱216K), and sales are down ≈ 15% from the Aug 2025 peak.

- Profitability is stable. Estimated margin is 29.3% (−0.2% YoY) and the gross-to-net gap stays at about ₱15–20K per month.
- Revenue is balanced. Channels split 33.8% Web / 33.4% Mobile App / 32.8% Marketplace; the top 3 categories (Toys, Electronics, Fashion) hold ≈ 40% of sales; the top five regions are within ≈ 7% of each other.
- Central Luzon is the #1 region (≈ ₱0.95M), ahead of Metro Manila (≈ ₱0.90M).
- Toys (≈ ₱0.75M) and Electronics (≈ ₱0.73M) are the top categories; Sports (≈ ₱0.58M) is the lowest.
- Order status mix: Completed 65.8%, Shipped 12%, Pending 10.2%, Cancelled 8.2%, Returned 3.8%, so about 22% of orders are not fulfilled.
- Customers buy about 2.5 times each at a ₱434 AOV.

Section 6: Insights

- Growth is not what the KPI strip suggests. Because of the unequal comparison windows, leadership could believe sales almost doubled when monthly sales are actually flat. AOV (+0.3%) and margin (−0.2%) are the only unaffected YoY KPIs. *Decision impact: targets and forecasts must not be based on the +92% figure.*
- The business has plateaued, with a softening late 2025. Without a growth driver, sales hover around ₱215K a month and have dropped since Aug 2025. Troughs around early 2025 suggest some seasonality, but this needs checking by category and channel.
- Margins are a strength; the lever is volume and basket size, not pricing. A steady ≈ 29% margin with consistent ≈ 7.4% discounting means profit moves with sales. Note that "estimated" margin excludes shipping, payment fees and refunds, so true margin is lower.
- There is no concentration risk, but also no clear growth engine. Even channel, category and regional splits protect against a single failure, but they also mean no segment is pulling ahead. Growth has to come from improving every segment or finding which one responds best to investment.
- Roughly one in five orders leaks revenue before fulfilment. Pending (≈ 10%), Cancelled (≈ 8%) and Returned (≈ 4%) orders are the largest recoverable pool. Even recovering a portion would raise sales without new customer acquisition.
- Repeat purchase is the biggest customer opportunity. At ≈ 2.5 orders per customer and a ₱434 AOV, increasing frequency or basket size has more impact than adding new customers alone. The Customers tab is built to show how loyalty tiers and order counts are distributed.
- The dashboard's own narrative needs hardening. Several "Key Insight" captions are static text that conflict with the charts (e.g. "online channels dominate", "wide margin" between categories, "metropolitan regions lead", "peaks in April and July"). Filters are also applied inconsistently between tabs. Fixing these is essential for the dashboard to be trusted by executives.

Section 7: Summary

The Executive Retail Intelligence Hub consolidates seven datasets into one cleaned, joined model and presents eight views of the business. Across Jan 2024 – Jan 2026 the business generated ₱5.37M gross and ₱4.97M net sales from 11,439 fulfilled orders and 4,506 customers, with an estimated profit of ₱1.46M (29.3% margin). Performance is stable and well-diversified across channels, categories and regions, but monthly sales are flat and have softened since Aug 2025. About 22% of orders do not reach fulfilment, and customers average only about 2.5 orders. The headline YoY growth of +92% is a calculation artifact, and some automated insight captions do not match the data.

Section 8: Conclusions

1. The business is stable and profitable but not growing. The data supports a plateau with a recent dip, not the near-doubling shown in the YoY KPIs.
2. Diversification is a strength. No channel, category or region dominates, so the risk of dependence on a single segment is low.
3. Revenue leakage and low repeat purchasing are the main opportunities. The ≈ 22% unfulfilled orders and ≈ 2.5 orders per customer offer more near-term upside than new acquisition alone.
4. The reporting layer needs correction before it is used for decisions. The YoY window logic, static captions, inconsistent filter scope and unstandardized region labels reduce confidence in the dashboard and should be fixed first.
5. Profitability figures are directional. Estimated margin ignores shipping, payment fees and refunds, so it should be treated as an upper bound.

Section 9: Recommendations

Now

- Fix the YoY comparison. For All Years, compare equal-length windows (e.g. last 12 months vs the 12 months before) and show "n/a" when the prior window lacks data. Apply the same fix to Unique Customers.
- Replace static captions with data-driven ones** generated from the filtered data (top category, share %, top region, peak month).
- Make filters consistent. Use one date scope for the KPIs, order status, payments and logistics, and include `selected_year` in *Reset Filters*.
- Standardize `region_name` in `load_data()` (trim and title-case, as done for categories) so `CALABARZON` and other labels cannot split or mismatch.

Next

- Investigate the Aug 2025 → Jan 2026 slowdown (≈ −15%) by category, channel and region to see whether it is seasonal or structural.
- Reduce the ≈ 22% non-fulfilled orders. Review Pending ageing (payment or stock issues), Cancelled reasons and Returned categories; target Pending → Shipped conversion first.
- Launch a repeat-purchase program (loyalty tier upgrades, post-purchase offers, bundles) aimed at lifting orders per customer above ≈ 2.5 and AOV above ₱434.
- Review discount effectiveness. Use the Profit & Discounts tab to find discount tiers that do not raise basket size; ≈ 7.4% of gross sales is given away.

- Run targeted growth tests in the lowest performers (CALABARZON, Sports, Groceries, Books) rather than only reinforcing leaders.

Later

- Move to true contribution margin by deducting shipping cost, payment gateway fees and refund losses (the data is already loaded in `payments`, `shipments` and `returns`).
- Add forecasting and targets: monthly sales forecast, target-vs-actual lines and anomaly flags for dips like early 2025.
- Add customer analytics: cohort retention, RFM segmentation and repeat rate by acquisition channel.
- Operationalize the data: scheduled refreshes, automated data-quality alerts (row-loss thresholds, unknown regions or categories) and a database or warehouse in place of the CSV folder lookup.
- Refactor the app: split the ≈ 4,200-line script into modules (`data.py`, `metrics.py`, one file per tab), remove duplicated filter/date blocks and cache heavy computations with `st.cache_data`.